

1.Determine if string halves are alike

```
public class Solution {
    public boolean halvesAreAlike(String s) {
        Set<Character> vowels = new HashSet<>();
        vowels.add('a'); vowels.add('e'); vowels.add('i'); vowels.add('o');
        vowels.add('u');
        vowels.add('A'); vowels.add('E'); vowels.add('I'); vowels.add('O');
        vowels.add('U');

        int vowelsCount = 0;
        int midIndex = s.length() / 2;

        for (int i = 0; i < midIndex; i++) {
            char charA = s.charAt(i);
            char charB = s.charAt(midIndex + i);
            if (vowels.contains(charA)) vowelsCount++;
            if (vowels.contains(charB)) vowelsCount--;
        }

        return vowelsCount == 0;
    }
}
```

2.Contains Duplicate

```
class Solution {
    public boolean containsDuplicate(int[] nums) {
        int n = nums.length;
        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                if (nums[i] == nums[j])
                    return true;
            }
        }
        return false;
    }
}
```

3.Move zeroes

```
class Solution {
    public void moveZeroes(int[] nums) {
```

```

        int k = 0;

        for(int num : nums){

            if(num != 0){
                nums[k++] = num;
            }
        }

        while(k < nums.length){
            nums[k++] = 0;
        }
    }
}

```

4. Transpose Matrix

```

class Solution {
    public int[][] transpose(int[][] matrix) {
        int row = matrix.length;
        int col = matrix[0].length;
        int[][] trans = new int[col][row];

        for (int i = 0; i < col; i++) {
            for (int j = 0; j < row; j++) {
                trans[i][j] = matrix[j][i];
            }
        }

        return trans;
    }
}

```

5. Matrix block sum

```

class Solution {
    public int[][] matrixBlockSum(int[][] mat, int k) {
        int[][] arr= new int[mat.length][mat[0].length];
        for(int i=0;i<mat.length;i++)
        {
            for(int j=0;j<mat[0].length;j++)
            {
                int sum=0;
                for(int r=Math.max(0,i-k);r<=i+k && r<mat.length;r++)
                {
                    for(int c=Math.max(j-k,0);c<=j+k && c<mat[0].length;c++)
                    {
                        sum= sum + mat[r][c];
                    }
                }
            }
        }
    }
}

```

```
        }
    }
    arr[i][j] = sum;
}
}
return arr;
}
```